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ADAPTIVE SERVOMECHANISM FOR PEAK-TO-AVERAGE POWER RATIO OPTIMIZATION IN FM DIGITAL RADIO TRANSMISSION

Abstract

A PAPR optimization method for digital FM radio transmission is disclosed. The method includes generating an Orthogonal Frequency-Division Multiplexed (OFDM) signal for transmission and applying an adaptive PAPR reduction algorithm to the generated signal. The method further includes determining a Modulation Error Ratio (MER) for one or more frequency sidebands and iteratively adjusting PAPR reduction parameters based on the measured MER values. The adjustment includes optimizing the number of PAPR reduction iterations for a highest-power primary digital sideband, optimizing QAM constellation constraint limits, and adjusting convergence bias weight settings for a lowest-power primary digital sideband and/or secondary digital sidebands to maintain target MER values. The method further includes updating PAPR reduction parameters in real-time until the measured MER values meet predefined target thresholds and executing instructions to coordinate the operation of an optimization servomechanism, ensuring dynamic PAPR optimization across multiple transmission modes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is related to and claims priority to U.S. Provisional Application No. 63/562,261, filed on Mar. 7, 2024, and titled “AUTOMATED PAR PARAMETER SETTING ALGORITHM”, which is hereby incorporated by reference in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure relates to the field of digital frequency modulation (FM) radio transmission, and particularly relates to managing Peak-to-Average Power Ratio (PAPR) in Orthogonal Frequency-Division Multiplexed (OFDM) signals used in digital FM broadcasting.

Description of the Related Art

[0003] With advancements in digital broadcasting, FM radio transmission has evolved from traditional analog techniques to more complex digital transmission schemes, such as OFDM-based modulation. These modern digital transmission systems offer higher spectral efficiency, improved resilience to interference, and better signal integrity over varying channel conditions. As a result, digital FM broadcasting has been widely adopted for delivering high-quality audio and data services. However, despite its advantages, OFDM-based digital FM transmission introduces significant challenges, particularly concerning Peak-to-Average Power Ratio (PAPR). The nature of OFDM signals results in high peak power levels compared to their average power levels, leading to increased nonlinear distortions in power amplifiers (PAs). This issue is further exacerbated when higher-order Quadrature Amplitude Modulation (QAM) schemes are used, which require stringent signal integrity for maintaining quality performance.

[0004] High PAPR in digital FM transmission systems presents several technical issues. One of the issues is that most power amplifiers (PAs) operate efficiently only within a limited power range, such that when the signal exhibits high peaks, the amplifier either clips the signal, introducing intermodulation distortions, or requires back-off operation, reducing power efficiency (i.e., power amplifier non-linearity). Another problem is that PAPR-induced distortions degrade the Modulation Error Ratio (MER), which is a critical performance parameter in digital radio transmission, and lower MER results in poor signal demodulation and reception quality (i.e., increased signal distortion). Yet another problem is that high PAPR leads to excessive spectral leakage, causing interference in adjacent frequency bands, which is undesirable in regulated FM broadcast environments (i.e., spectral regrowth and adjacent channel interference). Yet another problem is that transmitters must operate at a reduced average power level to accommodate peak values, leading to suboptimal power consumption and reduced transmission range (i.e., inefficient power utilization).

[0005] To mitigate these challenges, various PAPR reduction techniques have been explored, including clipping and filtering, coding techniques, and signal processing approaches. However, existing solutions either introduce significant signal distortion, increase computational complexity, or fail to adapt dynamically to varying transmission conditions. Thus, there remains a need for an

improved solution that effectively reduces PAPR while maintaining signal integrity, optimizing power efficiency, and ensuring compliance with transmission standards in digital FM broadcasting, and overcomes the above-mentioned drawbacks.

BRIEF SUMMARY

[0006] One or more embodiments are directed to a Peak-to-Average Power Ratio (PAPR) optimization system and method (hereinafter may also be termed “mechanism”) for digital Frequency Modulation (FM) radio transmission. The disclosed mechanism in the present disclosure includes an Orthogonal Frequency-Division Multiplexed (OFDM) modulator configured to generate the OFDM signal for transmission. A PAPR reduction module applies an adaptive peak-to-average power ratio reduction algorithm to the generated signal, ensuring efficient signal processing. The mechanism further comprises a measurement module that determines the Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal, facilitating real-time signal evaluation. To dynamically adjust and optimize transmission parameters, the mechanism incorporates an optimization servomechanism, which iteratively fine-tunes PAPR reduction parameters based on measured MER values. The optimization servomechanism includes a PAPR reduction iteration controller, which optimizes the number of PAPR reduction iterations for primary digital sidebands, and a QAM constraint controller, which refines constellation constraints for modulated signals. Additionally, a convergence bias controller dynamically adjusts bias weight settings for secondary digital sidebands and/or a lower-powered primary digital sideband to maintain target MER values. The mechanism also integrates a feedback loop module, which continuously updates PAPR reduction parameters in real-time until the measured MER values align with predefined target thresholds.

[0007] Further, the mechanism employs a control processor to coordinate the operation of the optimization servomechanism and maintain dynamic PAPR optimization across multiple transmission modes. The control processor executes a multi-phase adaptive optimization algorithm, systematically adjusting PAPR reduction parameters in distinct optimization stages to refine the signal quality. The mechanism is capable of adjusting transmission parameters dynamically based on real-time transmission conditions, ensuring efficient power utilization, signal integrity, and optimal digital FM broadcasting performance.

[0008] An embodiment of the present disclosure discloses the PAPR optimization system for digital FM radio transmission. The system includes an Orthogonal Frequency-Division Multiplexed (OFDM) modulator to generate the OFDM signal for transmission. The OFDM modulator plays a crucial role in ensuring efficient frequency-domain signal representation, enabling the application of PAPR reduction techniques. Further, the PAPR optimization system includes a PAPR reduction module to apply an adaptive peak-to-average power ratio reduction algorithm to the generated OFDM signal. The PAPR reduction module ensures that the PAPR reduction is effectively performed before transmission, thereby mitigating nonlinear distortions that may otherwise arise during amplification. The PAPR reduction module operates in conjunction with real-time measurements and feedback to dynamically optimize the PAPR levels.

[0009] In an embodiment, the PAPR optimization system includes a measurement module to determine the Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal. By continuously evaluating MER values, the measurement module assesses the impact of PAPR reduction on signal quality. The measured MER values serve as a critical metric for dynamically adjusting transmission parameters, ensuring that the optimized signal maintains high fidelity and low distortion.

[0010] In an embodiment, the PAPR optimization system also includes an optimization servomechanism to iteratively adjust one or more PAPR reduction parameters based on the measured MER values. The optimization servomechanism includes a PAPR reduction iteration controller, a QAM constraint controller, and a convergence bias controller, each configured to optimize different aspects of PAPR reduction. The PAPR reduction iteration controller is

responsible for optimizing the number of PAPR reduction iterations specifically for a primary digital sideband, ensuring efficient transmission with minimal peak power levels. The QAM constraint controller manages the QAM constellation constraint limits for modulated signals within the OFDM transmission, thereby preventing excessive distortion in higher-order modulation schemes such as 16-QAM and 64-QAM. The convergence bias controller fine-tunes bias weight settings for a lowest-power primary digital sideband and/or secondary digital sidebands, ensuring that lower-power sidebands maintain a stable and predictable signal structure. The optimization servomechanism operates continuously, refining the transmission parameters dynamically to maintain signal integrity.

[0011] In an embodiment, the PAPR optimization system also includes a feedback loop module to ensure continuous real-time updates of PAPR reduction parameters. The feedback loop module actively monitors and compares measured MER values against predefined target thresholds, dynamically adjusting parameters to achieve the best possible transmission quality. The feedback loop module is designed to prevent excessive iterations by stopping the optimization process when the measured MER values exceed the predefined target threshold, ensuring efficient convergence and preventing unnecessary computational overhead.

[0012] In an embodiment, the PAPR optimization system also includes a control processor in the PAPR optimization system to execute instructions for coordinating the operation of the optimization servomechanism and maintaining dynamic PAPR optimization across multiple transmission modes. The control processor is responsible for executing a multi-phase adaptive optimization algorithm, ensuring that different stages of PAPR optimization are conducted systematically. The optimization occurs in multiple phases, with the PAPR reduction iteration controller optimizing MER for the highest-power primary digital sideband in the first phase, the QAM constraint controller optimizing MER for QAM-modulated signals in the second phase, and the convergence bias controller fine-tuning bias weight adjustments to optimize MER for a lowest-power primary digital sideband and secondary digital sidebands in the third phase. The multi-phase adaptive optimization algorithm is capable of repeating in multiple stages, with each subsequent stage using smaller parameter increment values to refine optimization further. Additionally, the control processor enables real-time adaptation of transmission parameters, allowing the PAPR optimization system to dynamically adjust based on changing transmission conditions, ensuring efficient performance across different FM digital broadcast modes.

[0013] An embodiment of the present disclosure discloses the PAPR optimization method for digital FM radio transmission. The method includes the steps of generating the OFDM signal for transmission to facilitate digital FM broadcasting. Further, the method includes the steps of applying an adaptive peak-to-average power ratio reduction algorithm to the generated OFDM signal, ensuring the signal is optimized before transmission. The method further includes the steps of determining a Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal, providing a metric for assessing the effect of PAPR reduction on signal quality for a given set of PAPR reduction parameters.

[0014] In an embodiment, the method further includes the steps of adjusting, iteratively, one or more PAPR reduction parameters based on the measured MER values. The adjustment includes the steps of optimizing the number of PAPR reduction iterations for a primary digital sideband, ensuring controlled signal peaks. Further, the method includes the steps of optimizing QAM constellation constraint limits for modulated signals within the OFDM signal, enhancing the integrity of the transmitted data. Additionally, the method includes the steps of adjusting bias weight settings for a lowest-power primary digital sideband and/or secondary digital sidebands to maintain target MER values, ensuring that lower-power sidebands are efficiently managed. The method further includes the steps of updating the PAPR reduction parameters in real-time until the measured MER values meet predefined target MER thresholds. Moreover, the method includes the steps of executing instructions for coordinating the operation of the optimization servomechanism

and maintaining dynamic PAPR optimization across multiple transmission modes.

[0015] In an embodiment, the method further includes the steps of incrementally adjusting the number of PAPR reduction iterations by a predefined step value until the MER measurement of the primary digital sideband reaches a target MER threshold. Additionally, the method includes the steps of dynamically adjusting iteration values based on hardware processing constraints to optimize power efficiency. The method further includes the steps of incrementally adjusting the distance from a QAM constellation point to its nearest boundary to improve signal integrity, preventing errors in high-order modulation schemes.

[0016] In an embodiment, the method further includes the steps for handling a lowest-power primary digital sideband and/or secondary digital sidebands and maintaining signal stability by dynamically adjusting bias weight settings. The method further includes the steps of performing bias weight adjustments using multi-stage incremental optimization, wherein each stage utilizes finer adjustment steps to improve accuracy, allowing for more refined control over the signal's power distribution. Further, the method includes the steps of continuously monitoring and comparing measured MER values against target thresholds and adjusting the PAPR reduction parameters in real-time to ensure signal optimization. The method further includes the steps of stopping iterative adjustments when the measured MER values exceed the predefined target threshold, ensuring efficient convergence of the optimization process.

[0017] In an embodiment, the method further includes the steps of executing a multi-phase adaptive optimization algorithm, wherein in a first phase, optimizing the MER of the highest-power primary digital sideband; in a second phase, optimizing MER for 16 and 64 QAM-modulated signals; and in a third phase, fine-tuning convergence bias weight adjustments to optimize MER for lower-power primary and secondary digital sidebands. The method further includes the steps of repeating the multi-phase optimization algorithm in multiple stages, with each subsequent stage using smaller parameter increment values to refine optimization, ensuring gradual fine-tuning of the signal parameters. Further, the method includes the steps of adjusting parameters based on real-time transmission conditions, enabling adaptive tuning across different FM digital broadcast modes, ensuring that the system remains responsive to varying channel conditions, and maintains optimal transmission performance.

[0018] The features and advantages of the subject matter here will become more apparent in light of the following detailed description of selected embodiments, as illustrated in the accompanying FIGUREs. As will be realized, the subject matter disclosed is capable of modifications in various respects, all without departing from the scope of the subject matter. Accordingly, the drawings and the description are to be regarded as illustrative in nature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] In the figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label with a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0020] FIG. 1 illustrates an exemplary environment having a Peak-to-Average Power Ratio (PAPR) optimization system for digital Frequency Modulation (FM) radio transmission, in accordance with an embodiment of the present disclosure.

[0021] FIG. 2A illustrates a block diagram of the PAPR optimization system for digital FM radio transmission, in accordance with an embodiment of the present disclosure.

[0022] FIG. 2B illustrates an operation of the PAPR optimization system for digital FM radio

transmission, in accordance with an embodiment of the present disclosure.

[0023] FIG. 3 is a flow chart of an operation of an optimization servomechanism, in accordance with an embodiment of the present disclosure.

[0024] FIGS. 4A-4B is a flow chart of an operation of initialization in the optimization servomechanism, in accordance with an embodiment of the present disclosure.

[0025] FIG. 5 is a flow chart of an operation of phase 1, in accordance with an embodiment of the present disclosure.

[0026] FIGS. 6A-6C are a flow chart of an operation of phase 2, in accordance with an embodiment of the present disclosure.

[0027] FIGS. 7A-7C are a flow chart of an operation of phase 3, in accordance with an embodiment of the present disclosure.

[0028] FIGS. 8A-8B is a flow chart of an operation of setting MER targets, in accordance with an embodiment of the present disclosure.

[0029] FIG. 9 is a flow chart of an operation of resetting in the optimization servomechanism, in accordance with an embodiment of the present disclosure.

[0030] FIG. 10 is a flow chart of a PAPR optimization method for digital FM radio transmission, in accordance with an embodiment of the present disclosure.

[0031] FIG. 11 illustrates an exemplary computer unit in which or with which embodiments of the present disclosure may be utilized.

[0032] Other features of embodiments of the present disclosure will be apparent from accompanying drawings and detailed description that follows.

DETAILED DESCRIPTION

[0033] Embodiments of the present disclosure include various steps, which will be described below. The steps may be performed by hardware components or may be embodied in machine-executable instructions, which may be used to cause a general-purpose or special-purpose processor programmed with the instructions to perform the steps. Alternatively, steps may be performed by a combination of hardware, software, firmware, and/or by human operators.

[0034] Embodiments of the present disclosure may be provided as a computer program product, which may include a machine-readable storage medium tangibly embodying thereon instructions, which may be used to program the computer (or other electronic devices) to perform a process. The machine-readable medium may include, but is not limited to, fixed (hard) drives, magnetic tape, optical disks, compact disc read-only memories (CD-ROMs), and magneto-optical disks, semiconductor memories, such as ROMs, PROMs, random access memories (RAMs), programmable read-only memories (PROMs), erasable PROMs (EPROMs), electrically erasable PROMs (EEPROMs), flash memory, magnetic or optical cards, or other types of media/machine-readable medium suitable for storing electronic instructions (e.g., computer programming code, such as software or firmware).

[0035] Various methods described herein may be practiced by combining one or more machine-readable storage media containing the code according to the present disclosure with appropriate standard computer hardware to execute the code contained therein. An apparatus for practicing various embodiments of the present disclosure may involve one or more computers (or one or more processors within the single computer) and storage systems containing or having network access to a computer program(s) coded in accordance with various methods described herein, and the method steps of the disclosure could be accomplished by modules, routines, subroutines, or subparts of a computer program product.

Terminology

[0036] Brief definitions of terms used throughout this application are given below.

[0037] The terms “connected” or “coupled”, and related terms are used in an operational sense and are not necessarily limited to a direct connection or coupling. Thus, for example, two devices may be coupled directly, or via one or more intermediary media or devices. As another example, devices

may be coupled in such a way that information can be passed there between, while not sharing any physical connection with one another. Based on the disclosure provided herein, one of ordinary skill in the art will appreciate a variety of ways in which connection or coupling exists in accordance with the aforementioned definition.

[0038] If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.

[0039] As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context dictates otherwise.

[0040] The phrases “in an embodiment,” “according to one embodiment,” and the like generally mean the particular feature, structure, or characteristic following the phrase is included in at least one embodiment of the present disclosure and may be included in more than one embodiment of the present disclosure. Importantly, such phrases do not necessarily refer to the same embodiment.

[0041] Exemplary embodiments will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments are shown. This disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. These embodiments are provided so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those of ordinary skill in the art. Moreover, all statements herein reciting embodiments of the disclosure, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future (i.e., any elements developed that perform the same function, regardless of structure).

[0042] Thus, for example, it will be appreciated by those of ordinary skill in the art that the diagrams, schematics, illustrations, and the like represent conceptual views or processes illustrating systems and methods embodying this disclosure. The functions of the various elements shown in the figures may be provided through the use of dedicated hardware as well as hardware capable of executing associated software. Similarly, any switches shown in the figures are conceptual only. Their function may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the entity implementing this disclosure. Those of ordinary skill in the art further understand that the exemplary hardware, software, processes, methods, and/or operating systems described herein are for illustrative purposes and, thus, are not intended to be limited to any particular named.

[0043] Embodiments of the present disclosure relate to a Peak-to-Average Power Ratio (PAPR) optimization system and method (hereinafter may also be termed “mechanism”) for digital Frequency Modulation (FM) radio transmission. The disclosed mechanism in the present disclosure includes an Orthogonal Frequency-Division Multiplexed (OFDM) modulator configured to generate the OFDM signal for transmission. A PAPR reduction module applies an adaptive peak-to-average power ratio reduction algorithm to the generated signal, ensuring efficient signal processing. The mechanism further comprises a measurement module that determines the Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal, facilitating real-time signal evaluation. To dynamically adjust and optimize transmission parameters, the mechanism incorporates an optimization servomechanism, which iteratively fine-tunes PAPR reduction parameters based on measured MER values. The optimization servomechanism includes a PAPR reduction iteration controller, which optimizes the number of PAPR reduction iterations for the highest-power primary digital sideband, and a QAM constraint controller, which refines constellation constraints for modulated signals. Additionally, a

convergence bias controller dynamically adjusts bias weight settings for the lowest-power primary digital sideband and/or secondary digital sidebands to maintain target MER values. The mechanism also integrates a feedback loop module, which continuously updates PAPR reduction parameters in real-time until the measured MER values align with predefined target thresholds.

[0044] Further, the mechanism employs a control processor to coordinate the operation of the optimization servomechanism and maintain dynamic PAPR optimization across multiple transmission modes. The control processor executes a multi-phase adaptive optimization algorithm, systematically adjusting PAPR reduction parameters in distinct optimization stages to refine the signal quality. The mechanism is capable of adjusting transmission parameters dynamically based on real-time transmission conditions, ensuring efficient power utilization, signal integrity, and optimal digital FM broadcasting performance.

[0045] FIG. 1 illustrates an exemplary environment **100** having a Peak-to-Average Power Ratio (PAPR) optimization system **106** for digital Frequency Modulation (FM) radio transmission, in accordance with an embodiment of the present disclosure.

[0046] In an embodiment, the exemplary environment **100** may include a user device **102**, a network infrastructure **104**, the PAPR optimization system **106**, a database **108**, and an FM broadcast output **110**. In an embodiment, the elements within environment **100** may be connected through wired or wireless communication channels, enabling efficient data transmission and processing. The user device **102** may be any computing device, including but not limited to, a personal computer, workstation, server, or broadcasting terminal. The user device **102** may interact with the PAPR optimization system **106** via the network infrastructure **104**. The network infrastructure **104** may include, but is not limited to, a wired or wireless communication network, cloud-based infrastructure, local network, or a combination thereof. Through the network infrastructure **104**, the user device **102** may transmit input data related to digital FM radio transmission to the PAPR optimization system **106**. The PAPR optimization system **106** may be responsible for processing digital FM signals and optimizing Peak-to-Average Power Ratio (PAPR). The PAPR optimization system **106** may include various components and functionalities, which may be described in more detail in FIG. 2. The PAPR optimization system **106** may dynamically analyze transmission parameters and apply optimization techniques before forwarding the processed signal for further broadcasting.

[0047] In an embodiment, the database **108** may store various data related to PAPR optimization, including but not limited to, historical transmission data, predefined threshold values, system configurations, and real-time processing parameters. The PAPR optimization system **106** may retrieve and update data within the database **108** as needed, ensuring adaptive and intelligent decision-making for digital FM transmission. The connection between the database **108** and the PAPR optimization system **106** may be local, cloud-based, or remotely accessible. As the PAPR optimization system **106** processes and optimizes the FM baseband signal, the processed baseband signal may be forwarded to the FM broadcast output **110**. The FM broadcast output **110** may represent a transmission system that delivers the processed signal to end-users. The FM broadcast output **110** may include, but is not limited to, FM transmission towers, antennas, satellite transmission systems, or other broadcasting equipment. The optimized FM signal may be transmitted efficiently with improved power efficiency and signal integrity, reducing distortion and maintaining compliance with transmission standards.

[0048] In an embodiment, the PAPR optimization system **106** may interact with various components to optimize and enhance digital FM radio transmission. The user device **102** may facilitate input transmission through the network infrastructure **104**, while the PAPR optimization system **106** performs PAPR optimization based on data stored within the database **108**. The optimized FM signal may be transmitted via the FM broadcast output **110** to reach the intended audience. The overall environment **100** may be implemented in various configurations and may operate across different transmission platforms, without departing from the scope of the disclosure.

[0049] FIG. 2A illustrates a block diagram **200A** of the PAPR optimization system **106** for digital FM radio transmission, in accordance with an embodiment of the present disclosure. FIG. 2B illustrates an operation **200B** of the PAPR optimization system **106** for digital FM radio transmission, in accordance with an embodiment of the present disclosure. For the sake of brevity, FIGS. 2A and 2B have been explained together.

[0050] In an embodiment, the PAPR optimization system **106** (hereinafter may also be termed as the system **106**) may include an Orthogonal Frequency-Division Multiplexed (OFDM) modulator **202**, a PAPR reduction module **204**, a measurement module **206**, an optimization servomechanism **208**, a feedback loop module **210**, and a control processor **212**. The OFDM modulator **202**, a PAPR reduction module **204**, a measurement module **206**, an optimization servomechanism **208**, a feedback loop module **210**, and a control processor **212** may be communicatively coupled to a memory and a processor of the PAPR optimization system **106**. The processor may control the operations of the OFDM modulator **202**, the PAPR reduction module **204**, the measurement module **206**, the optimization servomechanism **208**, the feedback loop module **210**, and the control processor **212**.

[0051] In an embodiment of the present invention, the processor and the memory may form a part of a chipset installed in the PAPR optimization system **106**. In another embodiment of the present invention, the memory may be implemented as a static memory or a dynamic memory. In an example, the memory may be internal to the PAPR optimization system **106**, such as an onside-based storage. In another example, the memory may be external to the PAPR optimization system **106**, such as cloud-based storage. Further, the processor may be implemented as one or more microprocessors microcomputers, microcomputers, digital signal processors, central processing units, state machines, logic circuitries, and/or any devices that manipulate signals based on operational instructions.

[0052] In an embodiment, the OFDM modulator **202** may generate the OFDM signal for transmission and may ensure efficient frequency-domain signal representation, enabling the application of PAPR reduction techniques. The OFDM modulator **202** may process input data, as shown by block **214**, by dividing the input data into multiple orthogonal subcarriers, thereby enhancing spectral efficiency and minimizing inter-symbol interference (ISI). Further, the OFDM modulator **202** may support various modulation schemes, including but not limited to Quadrature Amplitude Modulation (QAM) and Phase Shift Keying (PSK), to accommodate different transmission requirements. Furthermore, the OFDM modulator **202** may operate in conjunction with channel coding mechanisms, such as forward error correction (FEC), to improve signal robustness against noise and interference. Moreover, the OFDM modulator **202** may perform adaptive modulation. Additionally, the OFDM modulator **202** may incorporate pilot signals to facilitate accurate channel estimation and equalization, ensuring reliable transmission. The generated OFDM signal may then be processed by the PAPR optimization system **106** to reduce peak power levels while maintaining signal integrity. In an embodiment, the OFDM modulator **202** may be implemented in software, hardware, or a combination thereof, and may be integrated within a transmitter system, a digital signal processing (DSP) unit, or a software-defined radio (SDR) platform. The functionality of the OFDM modulator **202** may be extended to support multiple transmission standards, including but not limited to Digital Audio Broadcasting (DAB), Digital Radio Mondiale (DRM), and other digital FM broadcasting protocols.

[0053] In an embodiment, the PAPR reduction module **204** may apply an adaptive peak-to-average power ratio reduction algorithm, as shown by block **216**, to the generated OFDM signal. Further, the PAPR reduction module **204** may analyze the signal's peak power levels relative to its average power levels and apply various PAPR reduction techniques to minimize signal distortion while maintaining transmission efficiency. The reduction techniques may include, but are not limited to clipping and filtering, tone reservation, selective mapping (SLM), partial transmit sequences (PTS), active constellation extension (ACE), and coding-based methods. In an embodiment, the PAPR

reduction module **204** may ensure that PAPR reduction is effectively performed before transmission, thereby mitigating nonlinear distortions that may otherwise arise during amplification. Further, the PAPR reduction module **204** may consider hardware constraints, such as power amplifier (PA) linearity and efficiency, ensuring that the signal remains within an acceptable range for high-power transmission. Furthermore, the PAPR reduction module **204** may implement adaptive thresholding to control the trade-off between PAPR reduction and bit error rate (BER) performance, ensuring optimal signal quality. In an embodiment, the PAPR reduction module **204** may operate in conjunction with real-time measurements and feedback to dynamically optimize PAPR levels. The dynamic optimization may include continuously monitoring the Modulation Error Ratio (MER), signal-to-noise ratio (SNR), and spectral regrowth characteristics. The PAPR reduction module **204** may adjust PAPR parameters adaptively, facilitating the system **106** to respond to varying transmission equipment conditions and impairments. Additionally, the PAPR reduction module **204** may interact with a control processor to refine optimization settings based on historical transmission data, thereby improving long-term transmission efficiency. Additionally, as shown by block servomechanism **218** in FIG. 2B, the PAPR reduction module **204** may integrate with the Measurement module **206** to allow MER measurements on the PAPR reduced signals. Furthermore, the PAPR reduction module **204** may integrate with the feedback loop module **210** to receive real-time adjustments to PAPR reduction parameters, as shown by block **226**, to improve transmission efficiency and signal quality.

[0054] In an embodiment, the measurement module **206** may determine the Modulation Error Ratio (MER), as shown by block **222**, for one or more frequency sidebands in the OFDM signal. The measurement module **206** may analyze the signal's modulation accuracy, providing real-time feedback on the quality and integrity of the transmitted signal. The MER values may represent signal quality and serve as a key indicator of signal fidelity and transmission efficiency. By continuously evaluating MER values, the measurement module **206** may assess the impact of PAPR reduction on signal quality and identify potential distortions or degradations introduced by the PAPR reduction module **204**. Furthermore, the PAPR reduction module **204** may support multi-band measurements. The multi-band measurements may include evaluating the MER separately for primary and secondary digital sidebands, ensuring a comprehensive assessment of transmission performance. The measured MER values may serve as a critical metric for dynamically adjusting transmission parameters and may ensure that the optimized signal maintains high fidelity and low distortion. In an embodiment, the measurement module **206** may integrate with the optimization servomechanism module **208** to compare measured MER values against predefined target thresholds, allowing for real-time adjustments to improve transmission efficiency.

[0055] In an embodiment, the optimization servomechanism **208** may adjust one or more PAPR reduction parameters based on the measured MER values. The optimization servomechanism **208** may function as an intelligent control unit within the PAPR optimization system **106**, continuously refining transmission parameters to achieve an optimal balance between signal integrity, power efficiency, and spectral performance. Further, the optimization servomechanism **208** may dynamically adapt to real-time transmission conditions, ensuring that PAPR reduction does not compromise modulation accuracy or overall transmission quality. Furthermore, the optimization servomechanism **208** may include a PAPR reduction iteration controller, a QAM constraint controller, and a convergence bias controller, each configured to optimize different aspects of PAPR reduction.

[0056] In an embodiment, the PAPR reduction iteration controller may be responsible for optimizing the number of PAPR reduction iterations specifically for a primary digital sideband, ensuring efficient transmission with minimal peak power levels. The PAR iteration controller may iteratively adjust the PAPR reduction iterations based on real-time MER feedback, preventing excessive processing overhead while maintaining a low distortion profile. Further, the PAR iteration controller may also implement adaptive iteration scaling, where iteration values are

dynamically adjusted based on signal characteristics and hardware constraints. The QAM constraint controller may manage the QAM constellation constraint limits for modulated signals within the OFDM transmission, thereby preventing excessive distortion in higher-order modulation schemes such as 16-QAM and 64-QAM. The controller may monitor signal degradation metrics, including EVM (Error Vector Magnitude) and constellation drift, ensuring that transmitted symbols remain within the permissible boundary constraints. Additionally, the QAM constraint controller may apply adaptive QAM scaling, where the distance between constellation points is dynamically modified to maintain optimal MER values under varying transmission conditions. The convergence bias controller may fine-tune bias weight settings for a lowest-power primary digital sideband and/or secondary digital sidebands, ensuring that lower-power sidebands maintain a stable and predictable signal structure. The controller may implement bias compensation techniques, dynamically adjusting power levels and weighting factors to prevent signal degradation in multi-band OFDM transmissions. The convergence bias controller may further interact with the feedback loop module **210** to optimize bias settings based on real-time transmission feedback, ensuring that a lowest-power primary digital sideband and/or secondary sidebands do not introduce unnecessary spectral regrowth or adjacent-channel interference.

[0057] In an embodiment, the optimization servomechanism **208** may operate continuously, refining the transmission parameters dynamically to maintain signal integrity. The optimization servomechanism **208** may also support multi-phase PAPR optimization, where sequential adjustments to PAPR reduction iteration values, QAM constraint limits, and bias weight settings are performed in multiple stages, allowing for a gradual and controlled optimization process.

[0058] In an embodiment, the feedback loop module **210** may ensure continuous real-time updates of PAPR reduction parameters, facilitating the PAPR optimization system **106** to dynamically adapt to changing signal conditions and transmission requirements. The feedback loop module **210** may operate as a closed-loop control mechanism, continuously refining the PAPR optimization process to maintain optimal signal integrity and power efficiency. The feedback loop module **210** may integrate with the measurement module **206** to retrieve Modulation Error Ratio (MER) values and compare them against predefined target thresholds, ensuring that the system **106** operates within the desired performance range. The feedback loop module **210** may actively monitor and compare measured MER values against predefined target thresholds, dynamically adjusting PAPR parameters, as shown by the block **226**, to achieve the best possible transmission quality. Additionally, the feedback loop module **210** may employ adaptive thresholding techniques, where MER targets are dynamically adjusted based on historical performance data and real-time transmission conditions. The dynamic adjustments may optimize the system **106** ability to maintain consistent and reliable signal performance.

[0059] In an embodiment, the feedback loop module **210** may prevent excessive iterations, through a loop filter as shown by block **224**, by stopping the optimization process when the measured MER values exceed the predefined target threshold, ensuring efficient convergence and preventing unnecessary computational overhead. The feedback loop module **210** may implement adaptive iteration control, where the system automatically adjusts the number of PAPR reduction iterations based on processing constraints and performance objectives. Furthermore, the feedback loop module **210** may work in conjunction with the optimization servomechanism **208** to refine PAPR reduction parameters in multiple stages, allowing for a gradual and controlled optimization process that minimizes signal distortion while maintaining power efficiency.

[0060] In an embodiment, the control processor **212** may execute instructions for coordinating the operation of the optimization servomechanism and maintaining dynamic PAPR optimization across multiple transmission modes. Further, the control processor **212** may function as the central processing unit of the PAPR optimization system **106**, managing real-time signal processing tasks, optimization workflows, and adaptive tuning mechanisms. The control processor **212** may interface with various system components, including the PAPR reduction module **204**, measurement module

206, feedback loop module **210**, and database **108** to ensure seamless operation and data-driven optimization. The control processor **212** may be responsible for executing a multi-phase adaptive optimization algorithm, ensuring that different stages of PAPR optimization are conducted systematically. The optimization may occur in multiple phases, with each phase targeting specific transmission parameters. In the first phase, the PAPR reduction iteration controller may optimize the highest-power primary digital sideband, ensuring that PAPR reduction iterations are adjusted to maintain efficient power levels without introducing excessive distortions. In a second phase, the QAM constraint controller may optimize the Modulation Error Ratio (MER) for QAM-modulated signals, ensuring that the constellation constraints remain within acceptable limits for high-order modulation schemes, such as 16-QAM and 64-QAM. In the third phase, the convergence bias controller may fine-tune bias weight adjustments for the lowest-power primary digital sideband and/or secondary digital sidebands, ensuring stable and predictable signal structure across multi-band OFDM transmission environments.

[0061] In an embodiment, the multi-phase adaptive optimization algorithm may be capable of repeating in multiple stages, with each subsequent stage using smaller parameter increment values to refine optimization further. The control processor **212** may incorporate machine learning-based adaptation or historical data analysis techniques, allowing the system **106** to progressively improve transmission quality over time. Additionally, the control processor **212** may manage resource allocation for different processing tasks, ensuring that real-time optimizations do not introduce unnecessary computational delays. Further, the control processor **212** may enable real-time adaptation of transmission parameters, allowing the PAPR optimization system **106** to dynamically adjust based on changing transmission conditions. This may include adapting to variations in channel conditions, power amplifier (such as a High-Power Amplifier (HPA), as shown by block **220**) linearity constraints, network congestion levels, and environmental factors that influence FM signal propagation via the FM broadcast output **110**. The control processor **212** may also integrate with cloud-based or remotely accessible data management systems, enabling remote monitoring, parameter tuning, and over-the-air (OTA) updates for continuous system improvements.

[0062] FIG. **3** is a flow chart **300** of an operation of the optimization servomechanism **208**, in accordance with an embodiment of the present disclosure. The process may begin at step **302**.

[0063] At step **304**, the operation of the optimization servomechanism **208** includes initializing system parameters to prepare the PAPR optimization system **106** for processing. The initialization may involve setting default values, retrieving stored optimization parameters, and establishing communication with system components. Further details regarding the initialization process will be described in the following paragraphs in conjunction with FIGS. **4A-4B**.

[0064] Once initialized, the process may proceed to a MER calculation at step **306**, where the measurement module **206** may determine the Modulation Error Ratio (MER) of the OFDM signal. The MER values may provide insight into signal integrity and modulation accuracy, forming the basis for subsequent PAPR reduction parameter adjustments. Following the MER calculation, the system may proceed to set MER targets, at step **308**, to define threshold values that the system aims to achieve during PAPR optimization. Such MER targets may be based on predefined system constraints, real-time transmission conditions, or adaptive learning models. A more detailed explanation of setting the MER target and its impact on optimization will be described in the following paragraphs in conjunction with FIGS. **8A-8B**.

[0065] Next, the optimization process may proceed through a multi-phase approach, i.e., phase 1 **310**, phase 2 **312**, and phase 3 **314**. Each phase may be responsible for adjusting specific PAPR reduction parameters and optimizing different aspects of the transmitted signal to meet the predefined MER targets. The first phase **310** may focus on initial parameter tuning such that an increase in PAPR reduction iterations causes a decrease in the measured MER, the second phase **312** may refine modulation-specific constraints such that an increase in QAM constraint limit causes an increase in measured MER, and the third phase **314** may perform final adjustments to

ensure transmission stability such that an increase in CB weight may cause an increase in measured MER. In an embodiment, once phase 1 is complete, phase 2 and phase 3 are performed in a similar manner. Further, when all three phases have been completed, the entire three-phase process is repeated for stage 2 using the smaller parameter increment values as described below. Furthermore, the process is flexible in that more than 2 stages can be incorporated with different increment values and target MERs. The specific functions and operations performed in each optimization phase will be discussed in more detail in conjunction with FIGS. 5, 6A-6C, and 7A-7C.

[0066] Next, to ensure a controlled and structured optimization process, the optimization servomechanism **208** may implement a stage-based approach **316**, where the process may iterate through multiple stages to fine-tune the optimization settings. Each iteration may refine PAPR reduction parameters incrementally until the system converges to an optimal state. After each iteration, the stage counter may be incremented, as shown by **320**, and the system may evaluate whether the maximum number of optimization stages (StageMax) has been reached, as shown by **318**. If additional optimization is required, the process may continue through another iteration cycle; otherwise, the process may proceed to completion.

[0067] If at any point the system **106** determines that further adjustments are required, a reset operation, as shown by **322**, may be performed to restore initial conditions before restarting the optimization process. The reset operation may ensure that the PAPR optimization system **106** does not converge to a suboptimal state and may be configured to selectively reset specific parameters while preserving historical optimization data. Further explanation of the reset operation and its impact on convergence stability will be detailed in more detail in conjunction with FIG. 9.

[0068] Upon completing the multi-stage optimization process, the system **106** may reach an end operation, as shown by **324**, where the final PAPR-optimized signal may be transmitted via the FM broadcast system.

[0069] FIGS. 4A-4B is a flow chart **400** of an operation of initialization in the optimization servomechanism, in accordance with an embodiment of the present disclosure. The initialization process may establish the initial conditions necessary for PAPR optimization and ensure that the system parameters are configured for effective operation. Further, the process may involve setting default values, defining threshold conditions, and determining system-specific configurations before proceeding to the MER calculation and iterative optimization phases.

[0070] In an embodiment, the initialization process may begin at block **402**, where system parameters may be set to default values. These parameters may include, but are not limited to, target thresholds, modulation constraints, and control variables for the optimization servomechanism **208**. The initialization process may then proceed to a decision block **404**, where the system may determine if $SSM=MSS$. If no, then the MER target reached values may be assigned for secondary upper and lower sidebands, as shown by block **406**. Or else, further conditional checks may be performed. Next, the system may determine, as shown by block **408**, if $PUSB>PLSB$. If yes, specific MER target reached values and weight parameters may be assigned accordingly, as shown by blocks **410** and **412**. Similarly, if $PUSB<PLSB$, as shown by block **414**, then a different set of MER target reached values and weight parameters may be applied, as shown by blocks **416** and **418**. If neither condition is met, a default assignment of MER target reached values may be performed, as shown by block **420**, ensuring that the system initializes properly regardless of power distribution conditions.

[0071] In an embodiment, the initialization process may continue at decision block **422**, where the system **106** may evaluate if $PSM=MP1X$. If yes, then the specific target MER values and MER target reached values may be assigned at block **424**. Otherwise, additional mode-specific evaluations may be performed at blocks **426**, **430**, and **434**, covering different modulation and transmission schemes such as MP1, MP2, MP3, MP11, MP5, MP6, and MP1XOV. Each of these modes may correspond to specific target MER values, as shown by blocks **428**, **432**, and **436**, ensuring that the system operates according to the transmission conditions and requirements. If

none of the transmission schemes evaluated in decision blocks **422**, **426**, **430**, or **434** were “Y”, the initialization process may further evaluate the Quadrature Amplitude Modulation (QAM) type at decision block **442**. If QAM type=16, then corresponding target MER values may be assigned at block **444**. Otherwise, higher target MER values may be applied at block **446**, ensuring that higher-order QAM schemes receive appropriate signal quality constraints.

[0072] In an embodiment, before proceeding to the MER calculation phase **306**, the system may perform an additional comparison of power levels between the upper and lower primary sidebands, as shown by block **438**. If necessary, a sideband lock condition may be set at block **440**, restricting further parameter modifications in subsequent phases. Once all initialization conditions have been met, the system may transition to the MER calculation step **306**, where the measurement module **206** may determine the actual MER values for further optimization.

[0073] FIG. **5** is a flow chart **500** of an operation of phase 1, in accordance with an embodiment of the present disclosure. In an embodiment, phase 1 **310** is responsible for optimizing PAPR reduction iterations to achieve the predefined target Modulation Error Ratio (MER) set during initialization. The phase 1 ensures that the PAPR optimization system **106** dynamically adjusts PAPR parameters based on real-time MER feedback before proceeding to subsequent optimization phases.

[0074] In an embodiment, the process begins by verifying whether Target 1 has been reached, as shown by block **502**. If the MER target condition has been satisfied, the process transitions directly to Phase 2 **312**. Otherwise, the system **106** proceeds with further iterative adjustments to optimize PAPR reduction iterations. In further optimization, the system determines whether the current MER value exceeds the target MER threshold, at decision block **504**. If the condition is met, the system **106** checks the MER direction flag (MERDIR1), at block **506**, to evaluate whether previous iterations have improved MER values as far as possible. If the MER direction flag indicates improvements are complete, it finalizes the target MER condition, as shown by block **514**, and proceeds back to Calculate MER **306**. Otherwise, the system **106** increments the total PAPR reduction iterations, as shown by block **508**, and changes the MER direction flag, allowing the PAPR reduction process to continue refining the signal if the PAPR reduction iterations have not reached MAX1. Next, the system **106** evaluates whether the PAPR reduction iterations have reached the maximum allowable limit (MAX1), as shown by block **510**. If the maximum limit has not been reached, the system exits the phase 1 loop to proceed back to **306**, otherwise, it sets the PAPR reduction iterations to MAX1 and finalizes the target MER condition, as shown by block **514**, and returns to **306**.

[0075] In an embodiment, if the condition at block **504** is not satisfied (i.e., MER does not exceed TARGET1MER), the system **106** proceeds to block **516**, where it decrements PAPR reduction iterations to refine the optimization process. The system **106** evaluates whether the PAR iteration<MIN1 at decision block **518**. If so, the process sets the PAPR reduction iteration count to its minimum allowable value (MIN1) and sets Target 1 as reached at block **520**. Then the system **106** also sets MERDIR1 to -1, as shown by block **522**, and proceeds back to Calculate MER **306**. Otherwise, the MER direction flag is evaluated, as shown by block **524**. If the MER direction flag is already -1, the MER target reached condition is finalized and the system **106** proceeds back to Calculate MER **306**. Otherwise, the system **106** sets MERDIR to -1 in block **522** and proceeds back to Calculate MER **306**.

[0076] FIGS. **6A-6C** are a flow chart **600** of an operation of phase 2, in accordance with an embodiment of the present disclosure.

[0077] In an embodiment, at first, the system evaluates whether Target 2 has been reached, at block **602**. If yes, then the process transitions to phase 3 **314** for further convergence bias weight optimization. If not, the system checks if Target 2 upper is already satisfied, at block **604**. Otherwise, the system **106** continues refining QAM constellation constraints based on the measured MER values. Next, at decision block **606**, the system **106** compares the upper sideband

MER (MER2U) against the predefined TARGET2MER threshold. If MER2U exceeds Target2MER, the system **106** decrements the upper-sideband QAM constellation limit (QAMCONLIMITU) by INC2, as shown by block **608**, and the MER direction flag (MERDir2U) is updated to indicate a downward adjustment. The system **106** then evaluates whether QAMCONLIMITU has dropped below the minimum threshold (MIN2), as shown by block **610**. If so, the system **106** assigns QAMCONLIMITU=MIN2 in block **612** and proceeds to block **614** to flag Target 2 upper as reached (TARGET2REACHEDU=1). Otherwise, if MER2U does not exceed Target2MER, then the process increments QAMCONLIMITU by INC2, as shown by block **616**, and proceeds to check if QAMCONLIMITU has exceeded the maximum limit (MAX2), as shown by block **618**. If yes, it is assigned QAMCONLU=MAX2, as shown by block **620**, ensuring it does not exceed predefined operating conditions. In addition, in box **620**, Target 2 upper is flagged as reached (TARGET2REACHEDU=1). Next, at block **622**, the system **106** sets the MER direction flag MERDir2U to 1 and proceeds to decision block **626** on FIG. **6B**. If the decision at block **618** is no, the process moves to a decision block **624** to check if MERDIR2U=-1. If yes, then the process moves to block **614** and sets TARGET2REACHEDU=1 and proceeds to decision block **626** on FIG. **6B**. If no, the process moves to block **622** and sets MIRD2U to 1 and proceeds to decision block **626** on FIG. **6B**.

[0078] In an embodiment, at decision block **626**, the system **106** checks if Target 2 has been reached for the lower sideband (Target2ReachedL=1). If no, MER2L is compared against Target2MER at block **628**. If MER2L exceeds Target2MER, the system decreases the lower-sideband QAM constellation limit (QAMCONLIMITL) by INC2, as shown by block **630**, and the MER direction flag (MERDIR2L) is updated to -1 to reflect this change. The process then verifies whether QAMCONLIMITL has fallen below the minimum threshold, as shown by block **632**. If not, system **106** proceeds to decision block **648** on FIG. **6C**. If so, QAMCONLIMITL is set to MIN2, as shown by block **634**, and Target 2 is flagged as reached, as shown by block **636**. The system **106** then proceeds to decision block **648** on FIG. **6C**. If the MER condition at block **628** is not satisfied, the system increments QAMCONLIMITL by INC2, as shown by block **638**, ensuring that MER optimization remains within acceptable bounds. The process then checks if QAMCONLIMITL has exceeded the predefined MAX2 threshold, as shown by block **640**. If so, QAMCONLIMITL is assigned MAX2, and Target 2 lower is flagged as reached, as shown by block **642**. Then MERDIR2L is set to 1 in block **644** and the system **106** proceeds to decision block **648** on FIG. **6C**. If additional refinements are necessary, the system **106** evaluates whether MERDIR2L=-1, as shown by block **646**. If so, Target 2 lower is set to reached, as shown by block **636**, and the system **106** then proceeds to decision block **648** on FIG. **6C**. If no, MERDIR2L is set to 1, and the system **106** then proceeds to decision block **648** on FIG. **6C**.

[0079] In an embodiment, at decision block **648**, the system **106** checks whether the sideband lock condition (SB_LOCK=1) has been set. If SB_LOCK is active, further modifications to QAM constellation constraints may be restricted, ensuring system stability. At block **650**, the system **106** assigns LIMIT based on the maximum value between QAMCONLIMITU and QAMCONLIMITL.

[0080] This ensures that both sidebands maintain synchronized constellation constraints, preventing imbalanced signal distortions. The finalized QAM limits are then assigned at block **652**, setting both QAMCONLIMITU and QAMCONLIMITL to the determined limit value. The system **106** then verifies whether Target 2 has been reached (as shown by blocks **654**, **656**, **660**, and **662**). If either Target2 upper or Target 2 lower target conditions have been satisfied, the other target 2, TARGET2REACHEDU or TARGET2REACHEDL, is flagged accordingly as shown in blocks **662** and **656**. Then Target 2 reached is set in box **658**. The system **106** then proceeds back to the Calculate MER block **306**. If SB_LOCK is not active, then the system **106** sets TARGET2REACHED if BOTH TARGET2REACHEDL and TARGET2REACHEDU are set, as shown in block **658** and decision blocks **664** and **666**.

[0081] FIGS. **7A-7C** are a flow chart **700** of an operation of phase 3, in accordance with an

embodiment of the present disclosure. Phase 3 **314** is responsible for fine-tuning convergence bias weight adjustments for lowest-power primary digital sideband and/or secondary digital sidebands, ensuring that lower-power sidebands maintain stability while minimizing distortion. Further, the phase 3 optimizes convergence bias weight parameters (CBWTP, CBWTSU, CBWTSL) based on real-time Modulation Error Ratio (MER) feedback, allowing for adaptive transmission parameter refinement before the final MER calculation step **306**.

[0082] In an embodiment, the process begins at block **702**, where the system **106** checks whether Target 3 has been reached (TARGET3REACHED=1). If Target 3 is already satisfied, the system **106** proceeds to finalize the optimization process. Otherwise, the system **106** proceeds to decision block **704** to check if Target 3 has already been reached for the lowest-power primary sideband. If not, the system **106** moves to decision block **706**, where the system **106** evaluates whether the measured MER3 value exceeds the target threshold (TARGET3MER). If the measured MER3 value does not exceed the target threshold, the system **106** increments the primary convergence bias weight parameter (CBWTP) by a predefined increment (INC3) at block **716**. This adjustment ensures that the bias weight compensation aligns with real-time signal conditions, maintaining transmission quality. The system **106** evaluates at decision block **718** whether CBWTP has exceeded the maximum allowable threshold (MAX3). If the limit is not exceeded, it moves on to decision block **724** to check if MERDIR3P=-1. If yes, it sets TARGET3REACHEDP to 1 as shown in block **714**. However, if the maximum allowable threshold is exceeded, the system **106** sets CBWTP to MAX3 and TARGET3REACHEDP to 1 at block **720**, and updates the MER direction flag (MERDIR3P=1) at block **722**, finalizing primary convergence bias weight adjustments. If the MER condition is not met at decision block **706**, the system **106** reduces CBWTP by INC3 and checks at decision block **710** whether it has fallen below the minimum threshold (MIN3). If so, the system **106** sets CBWTP=MIN3 at block **712** and confirms that Target 3 has been reached at block **714** for the primary sideband.

[0083] Following primary-sideband convergence bias weight adjustments, the system **106** proceeds to optimize the secondary upper-sideband convergence bias weight settings. At decision block **726**, the system **106** verifies whether TARGET3REACHEDSU=1. If the condition is met, it proceeds to decision block **748** on FIG. 7C. If MER3SU exceeds TARGET3MERS, the system **106** decrements the upper-sideband bias weight (CBWTSU) at block **730**. At decision block **732**, the system **106** checks whether CBWTSU is below the minimum threshold (MIN3). If so, it sets CBWTSU=MIN3 at block **734**, and sets TARGET3REACHEDSU to 1, as shown in block **736**. At decision block **728**, if MER3SU does not exceed TARGET3MERS, the system **106** increases CBWTSU by INC3, as shown at block **738**. Further, if the convergence bias weight exceeds MAX3 at decision block **740**, the system **106** assigns MAX3 at block **742** and flags Target 3 as reached for the upper secondary sideband. It then proceeds to block **744** and sets MERDIR3SU to 1, and advances to block **748** on FIG. 7C. If it does not exceed MAX3, it proceeds to decision block **746** to check if MERDIRSU=-1. If not, it sets MERDIRSU=1 at block **744**. If yes, it sets TARGET3REACHEDSU to 1 at block **736**. Once all adjustments are complete, the system **106** proceeds to secondary lower sideband optimization at block **748** on FIG. 7C.

[0084] At decision block **748**, the system **106** verifies whether Target 3 has been reached for the secondary lower sideband (TARGET3REACHEDSL=1). If not, the MER3SL value is compared against TARGET3MERS at block **756**. If the measured MER value exceeds the target, the secondary lower sideband convergence bias weight (CBWTSL) is decreased by INC3 at block **758**, and MERDIR3L is set to -1. The system then evaluates at decision block **760** whether CBWTSL is below MIN3. If so, it sets CBWTSL=MIN3 at block **762** to prevent excessive convergence bias reduction and sets the secondary lower target 3 reached as 1, as shown by block **764**. At block **756**, if the measured MER value does not exceed the target, the system **106** increases CBWTSL by INC3 at block **766**, to improve stability and keep the signal clear. If the convergence bias weight has reached MAX3 at decision block **768**, the system assigns MAX3 at block **770** and confirms

that the secondary lower Target 3 has been reached at block **764**. If the conditions are not met, the MERDIR3L direction flag is compared to -1 at block **772**. If no, MERDIR3SL is set to 1 and system **106** proceeds back to Calculate MER block **306**. If yes, TARGET3REACHEDSL is set to 1 at block **764**. If TARGET3REACHEDSL has been set to 1, the system checks at decision blocks **750** and **752** whether Target 3 has been reached for the primary sideband and the secondary upper sideband as well. If confirmed, TARGET3REACHED is set to 1 at block **754**. The process then transitions to MER calculation step **306**.

[0085] FIGS. **8A-8B** is a flow chart **800** of an operation of setting MER targets, in accordance with an embodiment of the present disclosure. In an embodiment, MER target setting is a critical step in the PAPR optimization process, as it determines the reference values that guide iterative adjustments throughout the optimization servomechanism. The system **106** considers sideband power levels, primary service mode (PSM), and modulation schemes to define the appropriate MER target values for the transmission process.

[0086] In an embodiment, the process begins at block **802** by checking if target 1 reaches 1. If no, then the process evaluates whether the upper-sideband power (PUSB) is greater than the lower-sideband power (PLSB) at decision block **804**. If this condition is met, the process proceeds to decision block **806**, where it verifies the Primary Service Mode (PSM) classification. If the PSM is categorized as MP1, MP2, MP3, MP5, MP6, MP11, or MP1X, the system **106** assigns MER1 to MER values corresponding to Upper Sideband Quadrature Phase Shift Keying (UQPSK) modulation, as shown by block **808**. Otherwise, the system **106** assigns MER1 to MER values corresponding to Upper Sideband QAM (UQAM) modulation as shown in block **810**. If PUSB is not less than PLSB, the system **106** proceeds to decision block **812**, where it checks whether PUSB is less than PLSB. If so, it applies a similar evaluation process, as shown by block **814**, for Lower sideband Quadrature Phase Shift Keying (LQPSK) and Lower sideband Quadrature Amplitude Modulation (QAM) at blocks **816** and **818**.

[0087] If neither condition is satisfied, the system **106** applies a similar evaluation process, as shown by block **820**, and calculates MER1 as an average of the QPSK and QAM values, as shown by blocks **822** and **824**, ensuring a balanced MER distribution for both sidebands. This guarantees that the transmission maintains consistent MER values across different modulation schemes before proceeding to the next stage. In an embodiment, at decision block **826**, the system **106** evaluates whether Target 2 has been reached (TARGET2REACHED=1). If the target has been satisfied, the system **106** proceeds to block **828** to set phase 3 targets. If not, the system **106** proceeds to the next steps of the phase 2 target setting process, as shown by blocks **846**, **848**, and **850**. At block **828**, the system **106** assigns MER targets for the secondary upper sideband (MER3SU) and the secondary lower sideband (MER3SL) based on previously determined values from the initial MER calculations (SMERDATU and SMERDATL, respectively). The system **106** then compares the primary upper and lower sideband power levels at decision blocks **830** and **838**. If PUSB is higher than PLSB, the system **106** sets the phase 3 MER target (MER3) to that of the upper secondary sideband based on modulation scheme (MERDATUQPSK or MERDATUQAM) and sets the primary sideband convergence bias target (CBWTP) to that of the lower primary sideband (CBWTL), as shown by blocks **832**, **834**, and **836**. If PUSB is lower than PLSB, the system **106** sets MER3 to that of the lower secondary sideband based on modulation scheme (MERDATLQPSK or MERDATLQAM) and sets the primary sideband convergence bias target to that of the upper primary sideband (CBWTU), as shown by blocks **840**, **842**, and **844**. The assigned MER targets are then propagated to the next optimization phase (i.e., Phase 1 at block **310**), ensuring that the system **106** dynamically refines PAPR reduction parameters based on these reference values.

[0088] FIG. **9** is a flow chart **900** of an operation of resetting in the optimization servomechanism, in accordance with an embodiment of the present disclosure. In an embodiment, the reset operation ensures that the PAPR optimization system **106** can reinitialize critical parameters and restore

specific values to a predefined baseline, allowing for controlled adaptation in subsequent optimization cycles. The reset process prepares the system **106** for recalculating MER values, ensuring that iterative adjustments proceed efficiently.

[0089] At block **902**, the reset process begins by initializing key system variables. This includes setting values such as SB_LOCK=0, TARGET1REACHED=0, TARGET2REACHED=0, TARGET3REACHED=0, and corresponding MER direction indicators (MERDIR1, MERDIR2U, MERDIR2L, MERDIR3P, MERDIR3SU, MERDIR3SL) to zero. Additionally, convergence bias weight parameters (CBWTINC) and QAM limit increments (QAMLIMITINC) are initialized to predefined values. This step ensures that the system has a clean state before continuing the next stage of PAPR optimization. Next, at decision block **904**, the system **106** evaluates whether the primary-upper-sideband power (PUSB) is greater than the primary-lower-sideband power (PLSB). If this condition is met, the system **106** assigns, at block **906**, Target 3 Upper as reached (TARGET3REACHEDU=1) and Target 2 Lower as reached (TARGET2REACHEDL=1), indicating that the current PAPR settings have achieved optimal MER values. At block **906** system **106** also sets Target 3 Primary Reached to Target 3 Lower Reached (TARGET3REACHEDP=TARGET3REACHEDL).

[0090] If PUSB is not greater than PLSB, the system **106** moves to decision block **916**, where it checks if PUSB is less than PLSB. If this condition holds, the system **106** assigns Target 3 Lower as reached, Target 2 Upper as reached, and sets Target 3 Primary reached to Target 3 Upper Reached, at block **918**. If neither condition is satisfied, the system assigns Target 3 as reached across all primary sidebands at block **920**, ensuring that primary sideband convergence bias weight adjustments are finalized before recalculating MER values. The system **106** then proceeds to decision block **908**, where it checks whether the Secondary Service Mode (SSM) is set to 5. If the condition is false, the system marks Target 3 as reached for both upper and lower secondary sidebands at block **914**, finalizing the Target 3 initializations. Otherwise, the system **106** moves to decision block **910**, where it evaluates whether QAMTYPE is greater than 4. If the condition is true, the system **106** resets Target 2 Reached values for the upper and lower sidebands at block **912**, ensuring that the MER recalibration accounts for additional adjustments required in the next iteration. Once all conditions have been evaluated, the system **106** transitions to MER calculation at block **306**, where the measurement module **206** determines the new MER values before resuming the next cycle of PAPR optimization.

[0091] FIG. **10** is a flow chart **1000** of a PAPR optimization method for digital FM radio transmission, in accordance with an embodiment of the present disclosure. The method starts at step **1002**.

[0092] At first, at step **1004**, the method may include the steps of generating the OFDM signal for transmission to facilitate digital FM broadcasting. Next, at step **1006** the method may include the steps of applying an adaptive peak-to-average power ratio reduction algorithm to the generated OFDM signal, ensuring the signal is optimized before transmission. Next, at step **1008**, the method may further include the steps of determining a Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal, providing a metric for assessing the effectiveness of PAPR reduction.

[0093] Next, at step **1010**, the method may further include the steps of adjusting, iteratively, one or more PAPR reduction parameters based on the measured MER values. The adjustment may include the steps of optimizing the number of PAPR reduction iterations for a primary digital sideband, ensuring controlled signal peaks. Further, the method may include the steps of optimizing QAM constellation constraint limits for modulated signals within the OFDM signal, enhancing the integrity of the transmitted data. Additionally, the method may include the steps of adjusting convergence bias weight settings for the lowest-power primary digital sideband and/or secondary digital sidebands to maintain target MER values, ensuring that lower-power sidebands are efficiently managed. Next, at step **1012**, the method may include the steps of updating the PAPR

reduction parameters in real-time until the measured MER values meet predefined target MER thresholds. Thereafter, at step **1014**, the method may include the steps of executing instructions for coordinating the operation of the optimization servomechanism and maintaining dynamic PAPR optimization across multiple transmission modes.

[0094] In an embodiment, the method may further include the steps of incrementally adjusting the number of PAPR iterations by a predefined step value until the primary digital sideband reaches a target MER threshold. Additionally, the method may include the steps of dynamically adjusting iteration values based on hardware processing constraints to optimize power efficiency. Further, the method may include the steps of optimizing modulation parameters for signals using 16-QAM and 64-QAM modulation schemes to minimize distortion. The method may further include the steps of incrementally adjusting the distance from a QAM constellation point to its nearest boundary to improve signal integrity, preventing errors in high-order modulation schemes.

[0095] In an embodiment, the method may further include the steps of prioritizing lower-power secondary digital sidebands and maintaining signal stability by dynamically adjusting bias weight settings. The method may further include the steps of performing bias weight adjustments using multi-stage incremental optimization, wherein each stage utilizes finer adjustment steps to improve accuracy, allowing for more refined control over the signal's power distribution. Further, the method may include the steps of continuously monitoring and comparing measured MER values against target thresholds and adjusting the PAPR parameters in real-time to ensure signal optimization. The method may further include the steps of stopping iterative adjustments when the measured MER values exceed the predefined target threshold, ensuring efficient convergence of the optimization process.

[0096] In an embodiment, the method may further include the steps of executing a multi-phase adaptive optimization algorithm, wherein in a first phase, optimizing the highest-power primary digital sideband; in a second phase, optimizing MER for QAM-modulated signals; and in a third phase, fine-tuning bias weight adjustments for secondary digital sidebands. The method may further include the steps of repeating the multi-phase optimization algorithm in multiple stages, with each subsequent stage using smaller parameter increment values to refine optimization, ensuring gradual fine-tuning of the signal parameters. Further, the method may include the steps of adjusting parameters based on real-time transmission conditions, enabling adaptive tuning across different FM digital broadcast modes, ensuring that the system remains responsive to varying channel conditions and maintains optimal transmission performance. The method ends at step **1016**.

[0097] FIG. **11** illustrates an exemplary computer unit **1100** in which or with which embodiments of the present disclosure may be utilized. As shown in FIG. **11**, a computer system **1100** includes an external storage device **1114**, a bus **1112**, a main memory **1106**, a read-only memory **1108**, a mass storage device **1110**, a communication port **1104**, and a processor **1102**.

[0098] Those skilled in the art will appreciate that computer system **1100** may include more than one processor **1102** and communication ports **1104**. Examples of processor **1102** include, but are not limited to, an Intel® Itanium® or Itanium 2 processor(s), or AMD® Opteron® or Athlon MP® processor(s), Motorola® lines of processors, FortiSOC™ system on chip processors or other future processors. The processor **1102** may include various modules associated with embodiments of the present disclosure.

[0099] The communication port **1104** can be any of an RS-232 port for use with a modem-based dialup connection, a 10/100 Ethernet port, a Gigabit or 10 Gigabit port using copper or fiber, a serial port, a parallel port, or other existing or future ports. The communication port **1104** may be chosen depending on a network, such as a Local Area Network (LAN), Wide Area Network (WAN), or any network to which the computer system connects.

[0100] The memory **1106** can be Random Access Memory (RAM), or any other dynamic storage device commonly known in the art. Read-Only Memory **1108** can be any static storage device(s)

e.g., but not limited to, a Programmable Read-Only Memory (PROM) chips for storing static information e.g., start-up or BIOS instructions for processor **1102**.

[0101] The mass storage **1110** may be any current or future mass storage solution, which can be used to store information and/or instructions. Exemplary mass storage solutions include, but are not limited to, Parallel Advanced Technology Attachment (PATA) or Serial Advanced Technology Attachment (SATA) hard disk drives or solid-state drives (internal or external, e.g., having Universal Serial Bus (USB) and/or Firewire interfaces), e.g. those available from Seagate (e.g., the Seagate Barracuda 7200 family) or Hitachi (e.g., the Hitachi Deskstar 7K1000), one or more optical discs, Redundant Array of Independent Disks (RAID) storage, e.g. an array of disks (e.g., SATA arrays), available from various vendors including Dot Hill Systems Corp., LaCie, Nexsan Technologies, Inc. and Enhance Technology, Inc.

[0102] The bus **1112** communicatively couples processor(s) **1102** with the other memory, storage, and communication blocks. The bus **1112** can be, e.g., a Peripheral Component Interconnect (PCI)/PCI Extended (PCI-X) bus, Small Computer System Interface (SCSI), USB, or the like, for connecting expansion cards, drives, and other subsystems as well as other buses, such a front side bus (FSB), which connects processor **1102** to a software system.

[0103] Optionally, operator and administrative interfaces, e.g., a display, keyboard, and a cursor control device, may also be coupled to bus **1112** to support direct operator interaction with the computer system. Other operator and administrative interfaces can be provided through network connections connected through communication port **1104**. An external storage device **1114** can be any kind of external hard-drives, floppy drives, IOMEGA® Zip Drives, Compact Disc-Read-Only Memory (CD-ROM), Compact Disc-Re-Writable (CD-RW), Digital Video Disk-Read Only Memory (DVD-ROM). The components described above are meant only to exemplify various possibilities. In no way should the aforementioned exemplary computer system limit the scope of the present disclosure.

[0104] While embodiments of the present disclosure have been illustrated and described, it will be clear that the disclosure is not limited to these embodiments only. Numerous modifications, changes, variations, substitutions, and equivalents will be apparent to those skilled in the art, without departing from the spirit and scope of the disclosure, as described in the claims.

[0105] Thus, it will be appreciated by those of ordinary skill in the art that the diagrams, schematics, illustrations, and the like represent conceptual views or processes illustrating systems and methods embodying this disclosure. The functions of the various elements shown in the figures may be provided through the use of dedicated hardware as well as hardware capable of executing associated software. Similarly, any switches shown in the figures are conceptual only. Their function may be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the entity implementing this disclosure. Those of ordinary skill in the art further understand that the exemplary hardware, software, processes, methods, and/or operating systems described herein are for illustrative purposes and, thus, are not intended to be limited to any particular named.

[0106] As used herein, and unless the context dictates otherwise, the term “coupled to” is intended to include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements). Therefore, the terms “coupled to” and “coupled with” are used synonymously. Within the context of this document terms “coupled to” and “coupled with” are also used euphemistically to mean “communicatively coupled with” over a network, where two or more devices can exchange data with each other over the network, possibly via one or more intermediary device.

[0107] It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims.

Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refer to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

[0108] While the foregoing describes various embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the invention is determined by the claims that follow. The invention is not limited to the described embodiments, versions, or examples, which are included to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art

Claims

1. A Peak-to-Average Power Ratio (PAPR) optimization system for digital Frequency Modulation (FM) radio transmission, the PAPR optimization system comprising: an Orthogonal Frequency-Division Multiplexed (OFDM) modulator configured to generate the OFDM signal for transmission; a PAPR reduction module configured to apply an adaptive peak-to-average power ratio reduction algorithm to the generated OFDM signal; a measurement module configured to determine a Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal; an optimization servomechanism configured to iteratively adjust one or more PAPR reduction parameters based on the measured MER values, wherein the optimization servomechanism further includes: a PAR iteration controller configured to optimize the number of PAPR reduction iterations for a primary digital sideband; a QAM constraint controller configured to optimize QAM constellation constraint limits for modulated signals within the OFDM signal; a convergence bias controller configured to adjust bias weight settings for at least one of: lowest-power primary digital sideband and secondary digital sidebands to maintain target MER values; a feedback loop module configured to update the PAPR reduction parameters in real-time until the measured MER values meet predefined target MER thresholds; and a control processor configured to execute instructions for coordinating the operation of the optimization servomechanism and maintaining dynamic PAPR optimization across multiple transmission modes.
2. The PAPR optimization system of claim 1, wherein the PAR iteration controller is configured to incrementally adjust the number of PAPR reduction iterations by a predefined step value until the highest-power primary digital sideband reaches a target MER threshold.
3. The PAPR optimization system of claim 2, wherein the PAR iteration controller dynamically adjusts iteration values based on hardware processing constraints to optimize power efficiency.
4. The PAPR optimization system of claim 1, wherein the QAM constraint controller is configured to optimize modulation parameters for signals using 16-QAM and 64-QAM modulation schemes to minimize distortion.
5. The PAPR optimization system of claim 4, wherein the QAM constraint controller incrementally adjusts the distance from a QAM constellation point to its nearest boundary to improve signal integrity.
6. The PAPR optimization system of claim 1, wherein the convergence bias controller is configured to prioritize at least one of: lowest-power primary digital sideband and secondary digital sidebands and maintain signal stability by dynamically adjusting convergence bias weight settings.
7. The PAPR optimization system of claim 6, wherein the convergence bias weight adjustments are performed using multi-stage incremental optimization, wherein each stage utilizes finer adjustment

steps to improve accuracy.

8. The PAPR optimization system of claim 1, wherein the feedback loop module is configured to continuously monitor and compare measured MER values against target thresholds and adjust the PAPR reduction parameters in real-time.

9. The PAPR optimization system of claim 8, wherein the feedback loop module stops iterative adjustments when the measured MER values exceed the predefined target threshold, ensuring efficient convergence.

10. The PAPR optimization system of claim 1, wherein the control processor is configured to execute a multi-phase adaptive optimization algorithm, wherein: in a first phase, the PAR iteration controller optimizes the highest-power primary digital sideband; in a second phase, the QAM constraint controller optimizes MER for QAM-modulated signals; and in a third phase, the convergence bias controller fine-tunes convergence bias weight adjustments for at least one of: lowest-power primary digital sideband and secondary digital sidebands.

11. The PAPR optimization system of claim 10, wherein the multi-phase optimization algorithm is configured to repeat in multiple stages, with each subsequent stage using smaller parameter increment values to refine optimization.

12. The PAPR optimization system of claim 1, wherein the control processor adjusts system parameters based on real-time transmission conditions, enabling adaptive tuning across different FM digital broadcast modes.

13. A Peak-to-Average Power Ratio (PAPR) optimization method for digital FM radio transmission, the PAPR optimization method comprising: generating the OFDM signal for transmission; applying an adaptive peak-to-average power ratio reduction algorithm to the generated OFDM signal; determining a Modulation Error Ratio (MER) for one or more frequency sidebands in the OFDM signal; adjusting, iteratively, one or more PAPR reduction parameters based on the measured MER values, wherein the adjusting further includes the steps of: optimizing the number of PAPR reduction iterations for a primary digital sideband; optimizing QAM constellation constraint limits for modulated signals within the OFDM signal; adjusting bias weight settings for at least one of: lowest-power primary digital sideband and secondary digital sidebands to maintain target MER values; updating the PAPR parameters in real-time until the measured MER values meet predefined target MER thresholds; and executing instructions for coordinating the operation of the optimization servomechanism and maintaining dynamic PAPR optimization across multiple transmission modes.

14. The PAPR optimization method of claim 13, further comprises: incrementally adjusting the number of PAPR reduction iterations by a predefined step value until the highest-power primary digital sideband reaches a target MER threshold; and dynamically adjusting iteration values based on hardware processing constraints to optimize power efficiency.

15. The PAPR optimization method of claim 13, further comprises: optimizing modulation parameters for signals using 16-QAM and 64-QAM modulation schemes to minimize distortion; and incrementally adjusting the distance from a QAM constellation point to its nearest boundary to improve signal integrity.

16. The PAPR optimization method of claim 13, further comprises prioritizing at least one of: lowest-power primary digital sideband and secondary digital sidebands and maintaining signal stability by dynamically adjusting convergence bias weight settings, wherein the convergence bias weight adjustments are performed using multi-stage incremental optimization, wherein each stage utilizes finer adjustment steps to improve accuracy.

17. The PAPR optimization method of claim 13, further comprises: continuously monitoring and comparing measured MER values against target thresholds to adjust the PAPR reduction parameters in real-time; and stopping iterative adjustments when the measured MER values exceed the predefined target threshold, ensuring efficient convergence.

18. The PAPR optimization method of claim 13, further comprises executing a multi-phase

adaptive optimization algorithm, wherein: in a first phase, optimizing the highest-power primary digital sideband; in a second phase, optimizing MER for QAM-modulated signals; and in a third phase, fine-tuning convergence bias weight adjustments for at least one of: lowest-power primary digital sideband and/or secondary digital sidebands.

19. The PAPR optimization method of claim 18, wherein the multi-phase optimization algorithm is configured to repeat in multiple stages, with each subsequent stage using smaller parameter increment values to refine optimization.

20. The PAPR optimization method of claim 13, further comprises adjusting parameters based on real-time transmission conditions, enabling adaptive tuning across different FM digital broadcast modes.
